FINC5001 FOUNDATION IN FINANCE
TOPIC 6 - THE CAPITAL ASSET PRICING MODEL (CAPM)

PRE-WORKSHOP QUESTIONS

Pre-workshop self-access questions set is designed for you to self-access your understanding of the lecture content. You are expected to complete these questions prior to the workshop. These questions will not be discussed in the workshop.

Pre-Q1

You would like to create a portfolio with the same level of systematic risk as the market, and you have $1,000,000 to invest. Given the following portfolios invested in various asset classes, complete the following table:

Asset Class	Investment	Beta
Income stocks	320,000	0.70
Large-cap stocks	250,000	1.10
Growth stocks		1.50
Treasury notes		

Pre-Q2

The following probability distribution relating to returns on an individual security and the market portfolio are given.

State	**Probability**	**Return on security (%)**	**Return on the market (%)**
Boom	0.10	32	25
Good	0.15	22	18
Average	0.30	10	14
Poor	0.25	3	6
Crash	0.20	-15	-5

I. What are the expected return and standard deviation of returns for both the security and the market portfolio?
II. What is the beta of both the security and the market portfolio?
III. What is the current risk-free rate of return?

Pre-Q3

Suppose there are two stocks in the market. Stock X has a beta of 0.5 while Stock Y has a beta of 1.5. If the risk-free rate is 6% and the market risk premium is 8%, what proportion do you need to invest in X and Y to achieve overall portfolio returns of 16%? What is the beta of the portfolio?

FINC5001

Question 1

You would like to create a portfolio with the same level of systematic risk as the market, and you have $1,000,000 to invest. Given the following portfolios invested in various asset classes, complete the following table:

Asset Class	Investment	Beta
Income stocks	320,000	0.70
Large-cap stocks	250,000	1.10
Growth stocks		1.50
Treasury notes		

Treasury notes are assumed to be a risk-free asset; and hence, have a beta of zero. You want a portfolio with the same amount of systematic risk as the market, which is a beta of one.

The beta of a portfolio is the weighted average of the betas of the stocks making up the portfolio:

$$\beta_{Portfolio} = w_1\beta_1 + w_2\beta_2 + \cdots + w_n\beta_n$$

$$1 = (0.32 \times 0.7) + (0.25 \times 1.1) + (w_{Growth\ stocks} \times 1.5) + (w_{Risk\ free} \times 0)$$

$$1 = 0.224 + 0.275 + (w_{Growth\ stocks} \times 1.5) + 0$$

$$\therefore w_{Growth\ stocks} = \frac{0.501}{1.5} = 0.334$$

$334,000 (= 1,000,000 × 0.334) is invested in the portfolio of growth stocks; therefore, leaving $96,000 (1,000,000 – 320,000 – 250,000 – 334,000) that will be invested in Treasury notes.

Question 2

The following probability distribution relating to returns on an individual security and the market portfolio are given.

State	Probability	Return on security (%)	Return on the market (%)
Boom	0.10	32	25
Good	0.15	22	18
Average	0.30	10	14
Poor	0.25	3	6
Crash	0.20	-15	-5

I. What are the expected return and standard deviation of returns for both the security and the market portfolio?

$$\bar{r}_S = \sum_{i=1}^{n} r_i P_i = (0.32)(0.10) + (0.22)(0.15) + (0.10)(0.30) + (0.03)(0.25) + (-0.15)(0.20)$$
$$= 0.0725 = 7.25\%$$

$$\bar{r}_M = \sum_{i=1}^{n} r_i P_i = (0.25)(0.10) + (0.18)(0.15) + (0.14)(0.30) + (0.06)(0.25) + (-0.05)(0.20)$$
$$= 0.0990 = 9.90\%$$

$$\sigma_s = \sqrt{\sum_{t=1}^{n} (r_t - \bar{r})^2 \cdot p(r)}$$

$$= \sqrt{\begin{array}{l}(0.32 - 0.0725)^2(0.1) + (0.22 - 0.0725)^2(0.15) + (0.1 - 0.0725)^2(0.3) + \\ (0.03 - 0.0725)^2(0.25) + (-0.15 - 0.0725)^2(0.2)\end{array}} = 14.13\%$$

$$\sigma_M = \sqrt{\sum_{t=1}^{n} (r_t - \bar{r})^2 \cdot p(r)}$$

$$= \sqrt{\begin{array}{l}(0.25 - 0.099)^2(0.1) + (0.18 - 0.099)^2(0.15) + (0.14 - 0.099)^2(0.3) + \\ (0.06 - 0.099)^2(0.25) + (-0.15 - 0.099)^2(0.2)\end{array}}$$

$$= 9.27\%$$

II. What is the beta of both the security and the market portfolio?

$$\begin{aligned}\sigma_{SM} &= \sum_{i=1}^{n} (r_{Si} - \bar{r}_S)(r_{Mi} - \bar{r}_M) \times P_i \\ &= (0.32 - 0.0725)(0.25 - 0.099)(0.10) + (0.22 - 0.0725)(0.18 - 0.099)(0.15) \\ &\quad + (0.10 - 0.0725)(0.14 - 0.099)(0.30) + (0.03 - 0.0725)(0.06 - 0.099)(0.25) \\ &\quad + (-0.15 - 0.0725)(-0.05 - 0.099)(0.20) \\ &= 0.012913\end{aligned}$$

$$\beta_S = \frac{\sigma_{SM}}{\sigma_M^2} = \frac{0.012913}{0.0927^2} = 1.50$$

$$\begin{aligned}\sigma_{MM} &= \sum_{i=1}^{n} (r_{Mi} - \bar{r}_M)(r_{Mi} - \bar{r}_M) \times P_i \\ &= (0.25 - 0.099)^2 (0.10) + (0.18 - 0.099)^2 (0.15) + (0.14 - 0.099)^2 (0.30) \\ &\quad + (0.06 - 0.099)^2 (0.25) + (-0.05 - 0.099)^2 (0.20) \\ &= 0.008589\end{aligned}$$

$$\beta_M = \frac{\sigma_{MM}}{\sigma_M^2} = \frac{0.008589}{0.0927^2} = 1.00$$

III. the current risk-free rate of return

$$E(r_i) = r_f + E(r_m - r_f)\beta_i$$

$$\therefore E(r_i) = r_f - r_f\beta_i + E(r_m)\beta_i$$

$$\therefore E(r_i) = r_f(1 - \beta_i) + E(r_m)\beta_i$$

$$\therefore r_f = \frac{E(r_i) - E(r_m)\beta_i}{(1 - \beta_i)} = \frac{0.0725 - (0.099)(1.50)}{1 - 1.50} = 0.0152 = 15.2\%$$

Question 3

Suppose there are two stocks in the market. Stock X has a beta of 0.5 while Stock Y has a beta of 1.5. If the risk-free rate is 6% and the market risk premium is 8%, what proportion do you need to invest in X and Y to achieve overall portfolio returns of 16%? What is the beta of the portfolio?

$$E(r_X) = r_f + E(r_m - r_f)\beta_X = 6 + (8)(0.5) = 10\%$$
$$E(r_Y) = r_f + E(r_m - r_f)\beta_Y = 6 + (8)(1.5) = 18\%$$

$$\begin{aligned} E(r_p) &= E(r_X)w_X + E(r_Y)w_Y \\ &= E(r_X)w_X + E(r_Y)(1 - w_X) \\ &= E(r_X)w_X + E(r_Y) - E(r_Y)w_X \\ &= w_X\left(E(r_X) - E(r_Y)\right) + E(r_Y) \end{aligned}$$

$$w_X = \frac{E(r_p) - E(r_Y)}{E(r_X) - E(r_Y)} = \frac{16-18}{10-18} = 0.25 = 25\%$$
$$w_Y = 1 - w_X = 1 - 0.25 = 0.75 = 75\%$$

$$\beta_p = \sum_{i=1}^{n} \beta_i w_i = \beta_X w_X + \beta_Y w_Y = (0.5)(0.25) + (1.5)(0.75) = 1.25$$